

Development of human resources for health information systems in the Americas

Regional workshop

Panama City, 11–15 September 2023

Meeting report

PAHO



Pan American
Health
Organization



World Health
Organization

Americas Region

Development of human resources for health information systems in the Americas

Regional workshop

Panama City, 11–15 September 2023

Meeting report

Washington, D.C., 2025

Development of human resources for health information systems in the Americas: Regional workshop. Panama City, 11–15 September 2023. Meeting report

PAHO/HSS/HR/25-0001

© **Pan American Health Organization, 2025**

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO license (CC BY-NC-SA 3.0 IGO).

Under the terms of this license, this work may be copied, redistributed, and adapted for non-commercial purposes, provided the new work is issued using the same or equivalent Creative Commons license and it is appropriately cited. In any use of this work, there should be no suggestion that the Pan American Health Organization (PAHO) endorses any specific organization, product, or service. Use of the PAHO logo is not permitted.

All reasonable precautions have been taken by PAHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall PAHO be liable for damages arising from its use.

Contents

Acknowledgments	vii
Abbreviations and acronyms	viii
Introduction	1
Part one. Overview of the regional workshop “Development of human resources for health information systems in the Americas”	3
General objective.....	3
Specific objectives	3
Topics covered	3
Participating countries and territories and organizations.....	3
Part two. Content and specific objectives of the regional workshop “Development of human resources for health information systems in the Americas”	5
Opening.....	5
Module 1. Global and regional overview of human resources for health information systems	7
Global agenda: National health workforce accounts.....	7
Regional perspective and priorities: Evidence, information, and knowledge in human resources for health	7
Regional overview: Data and information on human resources for health	8
Module 2. Experiences with the development of human resources for health information systems in countries of the Region of the Americas.....	8
Brazil: Universal health system and health workforce information system	8
Cuba: Registry of health professionals and technicians	9
Barbados: Human resources information system	9
Canada: Human resources for health information systems	9
Haiti: Development of the human resources for health information system in the Americas.....	10
Module 3. Overview of the national health workforce accounts.....	10
National health workforce accounts, mapping of data sources and actors, and measurement methodology	11
Practical exercise: Critical issues.....	11

Module 4. Methodology for occupation mapping in the development of human resources for health information systems	12
Chile: Mapping occupations in the health system	12
Costa Rica: Mapping occupations in the health system	13
Jamaica: Mapping occupations in the health system	13
Suriname: Mapping occupations in the health system.....	14
Mapping of occupations at the regional level	14
Practical exercise: Occupation mapping	15
Module 5. Methodology for mapping of actors in the development of human resources for health information systems.....	16
Belize: Mapping actors in the human resources for health system.....	16
El Salvador: National health workforce accounts for 2022	17
Paraguay: Integrated national human resources for health information system: Features, processes, and progress	17
Regional mapping of actors	17
Practical exercise: Mapping of actors	18
Module 6. CARICOM's human resources for health caribbean commission	18
Human resources for health caribbean commission	18
From the action task force to the human resources for health caribbean commission: An overview	19
Basic human resources for health indicators in the Caribbean	19
Module 7. Human resources for health information system maturity assessment.....	20
Information systems for health: Conceptual framework, information systems for health maturity model, and development of human resources for health information systems in the Americas	20
Bahamas: National committee on human resources and health workers	22
Panama: The human resources for health information system: A key pillar in health system development and policy-building	22
Uruguay: Maturity assessment of the information system	22
Presentation of the proposed maturity model for the human resources for health information system at the regional level.....	23
Practical exercise: Maturity model for human resources for health information systems	23

Challenges and priorities for Latin American countries	26
Bolivia (Plurinational State of).....	26
Chile	26
Costa Rica.....	26
Cuba	27
Dominican Republic	27
Ecuador	27
Guatemala	27
Panama.....	27
Paraguay.....	28
Peru	28
Uruguay	28
Challenges and priorities for Caribbean countries and territories and Canada.....	30
Anguilla	30
Antigua and Barbuda	30
Bahamas.....	30
Barbados.....	30
Belize.....	30
British Virgin Islands	30
Canada	30
Dominica.....	31
Grenada.....	31
Guyana	31
Jamaica	31
Montserrat	31
Saint Kitts and Nevis.....	31
Saint Lucia	31
Saint Vincent and the Grenadines.....	32
Suriname.....	32
Trinidad and Tobago	32
Turks and Caicos Islands.....	32

Part four. Results and conclusions

36

A brief summary of the workshop.....	36
Results	36
Conclusions.....	36
Recommendations.....	37

Annexes

38

Annex 1. Regional workshop agenda: “Development of human resources for health information systems in the Americas” for Latin American countries, 11–13 September 2023	38
Annex 2. Agenda of the regional workshop “Development of human resources for health information systems in the Americas” for English-speaking Caribbean countries and territories, Canada, and the United States, 13–15 September 2023	41
Annex 3. Representatives of the various focal points at the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023	44

Figures

1 Participants in the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023.	4
2 Opening of the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023	6
3 Participants in the regional workshop “Development of human resources for health information systems in the Americas” during the group practical exercise, Panama, 2023.	12
4 Group activity during the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023.	15
5 Presentation of the IS4H Maturity Model at the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023.	21
6 Screenshot of the tool designed to assess the maturity of human resources for health information systems, Panama, 2023	24
7 Participants in the regional workshop “Development of human resources for health information systems in the Americas” during group work, Panama, 2023.	25

Tables

1 Workshop modules and their link to results	6
2 Summary of technical cooperation needs and requirements in human resources for health in Latin American countries, in order of priority, 2023	28
3 Summary of technical cooperation needs and requirements in human resources for health in the Caribbean and Canada, in order of priority, 2023	33

Acknowledgments

This document was prepared by the Human Resources for Health Unit of the Department of Health Systems and Services of the Pan American Health Organization (PAHO), under the direction of Benjamín Puertas. This effort was coordinated by Gisele Almeida, PAHO Senior Advisor in Human Resources for Health Evidence and Knowledge, with the support of Annella Auer, Joel Caraballo, Ana Paula Cavalcante de Oliveira, Isabel Duré, and Mariana Lopes Galante, all from PAHO.

Abbreviations and acronyms

ABS	General Bureau of Statistics of Suriname (Dutch acronym)
CARICOM	Caribbean Community
CIHI	Canadian Institute for Health Information
COVID-19	coronavirus disease 2019
FISO	Suriname's Government Functional Information System (Dutch acronym)
HLMA	health labor market analysis
HRH	human resources for health
HRH-ATF	Human Resources for Health Action Task Force
HRHIS	human resources for health information system(s)
HSS/HR	Evidence and Information Team of the Human Resources for Health Unit (Health Systems and Services Department)
HWF	health workforce
ILO	International Labour Organization
IS4H	Information Systems for Health
IS4H-MM	Information Systems for Health Maturity Model
ISCO	International Standard Classification of Occupations
ISCO-08	International Standard Classification of Occupations 2008
JaHRVis	Jamaican Health Care Reporting and Visualization System
MINSAP	Cuban Ministry of Public Health
NHWA	National Health Workforce Accounts
PAHO	Pan American Health Organization
PHC	primary health care
SE-COMISCA	Executive Secretariat of the Council of Ministers of Health of Central America and the Dominican Republic
SINARHUS	Costa Rican National Human Resources for Health Information System
VCPH	Virtual Campus for Public Health
WHO	World Health Organization

Introduction

Human resources for health information systems (HRHIS) are a key tool for the effective management of human resources for health (HRH) and for supporting the implementation of policies based on information and evidence. However, many countries in the Region of the Americas face challenges in establishing an integrated HRHIS that enables effective monitoring of the health labor market. The lack of timely, accurate, and relevant information for management creates limitations in analyzing and understanding the training, size, composition, qualifications, and distribution of the health workforce – both at the national and subnational levels – as well as in planning within this scope.

At the same time, an increasing amount of health sector data stored in various fragmented records and sources hinders the development and governance of information systems. Greater internal coordination and integration with the National Health Workforce Accounts (NHWA) platform could enhance programmatic coordination among countries and promote standardized practices.

The work carried out by the Human Resources for Health Unit of the Pan American Health Organization (PAHO/HSS/HR) since 2021 has helped identify key areas for technical cooperation to support countries in developing their information systems and fulfilling their regional and global commitments. First, assessing the maturity levels of each system will help guide cooperation efforts at the national level. Additionally, in a highly intersectoral and fragmented field such as the health workforce, data are managed by various governmental actors, including national health and health service authorities, the private sector, social security institutions, professional associations, and other country-specific entities. Gaining a comprehensive understanding of this network of actors will facilitate the identification of strategies to establish effective partnerships that promote information exchange.

To effectively manage the health workforce and improve information monitoring, it is essential to conduct a health labor market analysis (HLMA). This involves identifying trends and providing a comprehensive understanding of the key factors affecting the supply and demand of health workers, in order to improve forecasting and planning for future needs and guide short-term strategies to address immediate issues. Therefore, when initiating an HLMA in a country, it is crucial to first assess the implementation of the NHWA. Depending on the situation, it may be advisable to strengthen the implementation of the NHWA, utilizing the network of involved actors.

At the global level, the work done to address the requirements of the NHWA has revealed particularities in some professions that do not easily fit into the categories of the International Standard Classification of Occupations (ISCO) of the International Labour Organization (ILO). The diversity of professions, titles, and regulations across countries requires a specific study in the Region to facilitate understanding and comparability.

The differences in HRHIS at the regional level offer opportunities to share best practices on these issues. Exchange among countries can enrich perspectives and options for building information systems. It was for this reason that the subregional workshop titled “Development of Human Resources for Health Information Systems in the Americas” (Image 1) was organized. The workshop’s aim was to address key topics for developing and consolidating HRHIS at national, regional, and global levels; promote the creation of a community of practice; and identify cooperative actions among countries.

This report consists of four parts that address the general aspects of the workshop, workshop contents, challenges, and priorities of the countries in the Region, as well as the results and conclusions of the workshop.

PART ONE

Overview of the regional workshop “Development of human resources for health information systems in the Americas”

General objective

To address key topics for the development and consolidation of HRHIS at the national level, and for their integration at the regional and global levels.

Specific objectives

- Establish a methodology for mapping health occupations in the Region;
- Identify the actors involved in the development of HRHIS in the countries of the Americas;
- Validate a methodology for assessing HRHIS maturity;
- Update the status of the NHTWA at the regional and global levels;
- Identify the concept and steps of an HLMA;
- Introduce the use of the NHTWA to inform HLMA in the context of the Americas.

Topics covered

- Global and regional HRHIS framework;
- NHTWA at the regional and global levels;
- Mapping of health occupations in the Region;
- Mapping of actors involved in the development of HRHIS in the countries of the Americas;
- HRHIS maturity assessment.

Participating countries and territories and organizations

Anguilla, Antigua and Barbuda, Bahamas, Barbados, Belize, Bolivia (Plurinational State of), Brazil, British Virgin Islands, Canada, Chile, Colombia, Costa Rica, Cuba, Dominica, Dominican Republic, Ecuador, El Salvador, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Montserrat, Nicaragua, Panama, Paraguay, Peru, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, Turks and Caicos Islands, Uruguay, Venezuela (Bolivarian Republic of), Executive Secretariat of the Council of Ministers of Health of Central America and the Dominican Republic (SE-COMISCA), and PAHO (for more information, see Annex 3).

IMAGE 1 Participants in the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023



© PAHO

PART TWO

Content and specific objectives of the regional workshop “Development of human resources for health information systems in the Americas”

This second part presents the workshop content in detail and addresses each of the specific objectives aimed at achieving the following results:

- **R1.** Analysis of the HRHIS situation at the regional and global levels;
- **R2.** Presentation of the NHTWA framework and exchange of experiences among countries;
- **R3.** Introduction to the methodology for mapping health occupations and practical exercises on its use;
- **R4.** Presentation of a proposed methodology for mapping actors involved in the development of HRHIS;
- **R5.** Presentation of a validated methodology for assessing HRHIS maturity.

To achieve these results, the workshop was divided into modules (Table 1).

The following sections describe the activities carried out in each module, as well as the participation of different countries and the work conducted at the regional and global levels. Each module included presentations, group activities, discussions, and exchanges.

To facilitate collaborative work and maintain a record of the sessions, a dedicated space was established within the “Strengthening Human Resources for Health Information Systems” Community of Practice, available on the PAHO Virtual Campus for Public Health (VCPH). This space contains the presentations and minutes from all workshop sessions: <https://campus.paho.org/mooc/course/view.php?id=290§ion=0>.

Opening

The opening of the workshop was led by PAHO Human Resources for Health Unit Chief, Benjamín Puertas; PAHO Advisor on Health Systems and Services in Panama, Leah-Mari Richards; and Head of the Human Resources Planning Department of the Ministry of Health of Panama, Dagoberto Vega (Image 2).

TABLE 1 Workshop modules and their link to results

Module	Name	Result
Opening	Introduction and Presentation	-
Module 1	Global and Regional Overview of Human Resources for Health Information Systems (HRHIS)	R1
Module 2	Experiences with the Development of HRHIS in Countries of the Region of the Americas	R1
Module 3	Overview of the National Health Workforce Accounts (NHWA)	R2
Module 4	Methodology for Occupation Mapping in the Development of HRHIS	R3
Module 5	Methodology for Mapping of actors in the Development of HRHIS	R4
Module 6	CARICOM's Human Resources for Health Caribbean Commission	R1
Module 7	Methodology for Assessing the Maturity of HRHIS	R5

Source: PAHO.

IMAGE 2 Opening of the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023

© PAHO

From the outset, emphasis was placed on the importance of HRHIS in improving decision-making and reducing gaps in HRH, as well as in the path toward resilient health systems. In this context, the objectives, agenda, methodology, and expected results were presented, highlighting the importance of the active participation of the countries.

Gisele Almeida, Advisor in Human Resources for Health Evidence and Knowledge at PAHO, Coordinator of the Evidence and Information Team of the Human Resources for Health Unit (HSS/HR) at PAHO, and workshop coordinator, introduced the country representatives, who signaled their presence with their respective flags.

Finally, a video was shared about the history of the founding of PAHO in the Americas and its relationship with Panama.

Module 1. Global and regional overview of human resources for health information systems

During this module, the HRHIS advisors from PAHO presented the regional and global overview through the following three presentations.

Global agenda: National health workforce accounts

Mathieu Boniol, Advisor in Statistics Management, Data, Evidence, and Knowledge from the World Health Organization (WHO) Health Workforce Department, provided an overview of the global and regional health workforce situation. He emphasized the importance of the NHTWA in aligning health workforce data with data management systems, fostering dialogue among countries, and supporting the WHO global policies on the subject.

Regarding the global situation of HRH, he outlined the current challenges, which include an estimated shortage of 20 million health workers worldwide, imbalances in staff competencies, massive migration of HRH, unequal geographical distribution, aging HRH, low productivity, insufficient education, and limited development of specific policies.

Speaking about the NHTWA, he highlighted the opportunity it offers for countries to progressively improve the availability, quality, and use of data through a set of indicators, which should be adapted to each country's needs. From this perspective, the NHTWA was presented as a key tool for better planning and decision-making in health systems. At the global level, it facilitates the monitoring of health worker mobility, linking with the *WHO global code of practice on the international recruitment of health personnel*.

Finally, he emphasized that addressing these challenges requires a systemic approach, based on multisectoral collaboration and the involvement of multiple actors.

Regional perspective and priorities: Evidence, information, and knowledge in human resources for health

Gisele Almeida, for her part, provided a regional perspective on the development of HRHIS. She highlighted that these systems represent one of the key functions for health governance, the design of HRH policies, and capacity-building in countries. She also emphasized that Latin America and the Caribbean face a chronic deficit of 600 000 health professionals, as well as an uneven distribution of HRH in the Region, which limits the prospects of achieving the Sustainable Development Goals by 2030. She identified common issues in the countries of the Region, such as the fragmentation of HRH data, limited access to information from the private sector, and the lack of coordination with other sectors – factors that hinder the collection of data on the distribution, mobility, training, and competencies of the health workforce.

Regarding the NHTWA, the speaker agreed with Mr. Boniol that it offers a valuable opportunity by providing methodological definitions that can be adapted to the needs and characteristics of the countries in the Region, which would help strengthen HRHIS. Therefore, it is crucial for countries to implement and strengthen their HRHIS

to ensure the availability and proper use of data in policy decision-making, while also addressing other challenges related to HRH management, such as the high turnover of qualified personnel.

Finally, she proposed a road map for the Region that includes several processes: i) HRHIS maturity assessment; ii) mapping of occupations to ensure accurate categorization of professions; iii) mapping of actors to identify the producers and owners of HRH data; and iv) a health labor market analysis to identify trends and key factors influencing the supply and demand of health workers.

Regional overview: Data and information on human resources for health

The members of HSS/HR (Isabel Duré, Ana Paula Cavalcante de Oliveira, and Mariana Lopes Galante) presented an analysis of the indicators available in the NHWA at the regional level, as well as the results of the 2023 survey conducted with the focal points who provided the data. They highlighted that the density of human resources for health in the Americas stands at 63.3 per 10 000 population, exceeding the global target of 44.5. However, significant gaps exist between countries, with 16 countries falling below the global target, while others, such as Cuba and the United States of America, raise the average. Regarding the availability of health personnel, the subregions with the greatest deficiencies are the Caribbean and Central America. As for the results of the survey of NHWA users, the main challenges identified included difficulties in collecting data, poor data quality, and access to information restricted to the public sector.

Next, they presented a proposal of 17 prioritized core indicators for monitoring in the Region, based on the feasibility of data collection and the regional commitments made by countries to improve the availability, quality, and distribution of HRH in the Region. After the approval of this set of indicators, the countries expressed a need for more training and additional specialized staff to strengthen HRHIS.

Module 2. Experiences with the development of human resources for health information systems in countries of the Region of the Americas

In this module, the regional perspective was enriched by the experiences shared by several countries on the development of their HRHIS. Brazil and Cuba were invited as countries of reference in Latin America due to their progress in HRH management; Barbados, Canada, and Haiti were invited to represent English-speaking and Caribbean countries. Below is a summary of each country's interventions.

Brazil: Universal health system and health workforce information system

Isabela Cardoso M. Pinto, Secretary of Health Workforce Management and Education in Brazil, highlighted the importance of the health workforce (HWF) for the functioning of the health system and emphasized the need to identify HRH demands based on population density and equitable access to healthcare. Her analysis included mapping the institutions involved in the development of the HWF and key aspects such as attraction, settling, retention, distribution, availability, training, and migration of HRH in the country. Brazil has nearly 3 million HRH registered in various databases, including those of the national health facility registry, professional associations, the national higher education census, different health regulatory entities, and organizations responsible for primary health care and hospital services. Of the registered workers, 75% are women; 49% are professionals, 37% are technicians, and the remainder are workers with basic education levels. Brazil's HRHIS allows for the analysis of the workforce composition in areas such as nursing, medicine, physiotherapy, pharmacy, and dentistry. Among the various tools used to update the information system are the national health workforce census and the national health workforce information center. Regarding the use of the data, Dr. Cardoso emphasized that the evidence provided by HRHIS has made it possible to establish territorial priorities, develop a national policy for the management of health education and labor, and make progress in organizing an international conference on the topic.

Cuba: Registry of health professionals and technicians

Marcos Agustín del Risco del Río, from the Human Resources in Health Unit, presented Cuba's health system, emphasizing its registry of health professionals and technicians as a robust HRH information system. Under the oversight and management of the Ministry of Public Health (MINSAP), this system integrates three key processes: training centers, a national registry of professionals managed by MINSAP, and work assignments in health establishments and networks. The main groups of registered professionals include those in medicine, dentistry, nursing, and technical roles. The system allows for the management of variables such as province, type of professional, qualification level, workplace, age, specialty, and gender, facilitating the measurement of the distribution of professionals, and other key aspects for decision-making. By 2022, Cuba had 94 066 registered medical professionals and 17 657 dentists, with a ratio of 118 people per medical professional and 627 people per dentist. Finally, Mr. del Risco emphasized the importance of the reports generated by the system, noting that their analyses are crucial for informing decision-making.

Barbados: Human resources information system

Stacie Goring, Director of Planning and Research at the Ministry of Health and Wellness of Barbados, explained that HRH data collection in the country is carried out using various sources, such as population censuses, workforce surveys, facility assessments, and administrative databases. Information is also gathered by the Accreditation Council, which includes demographic data such as age, gender, and health professionals' qualifications.

However, data collection in Barbados primarily focuses on clinical decision-making and patient information, with limited emphasis on HRH-related data. Additionally, data management remains a paper-based system, which poses challenges for quality control. Efforts are currently underway to digitize the registration process.

During the COVID-19 pandemic, a special unit was established to manage the additional human resources recruited. This unit created a database to efficiently deploy personnel during the crisis, but it was later dissolved.

Barbados faces challenges related to system fragmentation, a lack of approved human resources management policies, and poor integration of various digital tools. Additionally, high staff turnover necessitates frequent training. Data analysis is conducted only on demand, limiting the monitoring of global indicators such as workforce migration.

Barbados is seeking technical assistance to improve data collection, cleaning, and validation, as well as training in managing data from multiple sources. It also requests support to address the challenge of linking human resources data with broader health information, enhancing interoperability among digital tools, and providing continuous training on the use of NHWA tools.

Canada: Human resources for health information systems

The representatives from Canada introduced the Canadian Institute for Health Information (CIHI), an organization primarily funded by Health Canada. CIHI is responsible for managing and standardizing health data across the country, ensuring national data comparability.

Canada's healthcare framework consists of 13 provinces and territories, each with independent healthcare systems, as well as a federal system that funds health services for Indigenous communities and the armed forces. This federated model results in independent data collection, with CIHI playing a key role in gathering and managing these data through linked databases that cover millions of patient records.

CIHI has a Centre of Excellence for health workforce data, aimed at improving data access and quality to support evidence-based planning. Its five-year vision includes developing a modernized, digitally accessible HRH data and information system. This system is designed to meet the needs of the healthcare system and address emerging health policy issues related to care models, demographic and epidemiological shifts, such as virtual care, equitable access, aging, and patient mental health.

One of the challenges Canada faces is the fragmentation of data across different jurisdictions and the need to establish comparable data standards nationwide. To overcome these challenges, investment in technical infrastructure and the development of methodologies such as the “National Unique Identifier” is required to enable accurate counts and a clearer understanding of workforce availability. The implementation of the full-time equivalent methodology is also crucial, as it measures workforce capacity more accurately by counting work hours rather than simply the number of employees, which could either underestimate or overestimate availability. This approach provides a more detailed view of actual capacity, facilitating planning and helping to measure workforce sustainability, thus preventing chain effects such as burnout and staff shortages.

Haiti: Development of the human resources for health information system in the Americas

Ketler Guillaume, Director of the Human Resources for Health Directorate at the Ministry of Public Health and Population of Haiti, gave a presentation on the country’s HRHIS, in which he explained that it is a digital system operated online, implemented in 2011 following a health workforce census. The system provides individual data on health personnel, facilitates administrative management, and allows for the generation of pre-programmed reports that can be exported in formats such as Excel.

Regarding HRHIS governance, the data are accessible to national statistics offices and professional associations, although there are currently no interoperability mechanisms with other national or regional systems. The creation of a multisectoral human resources for health committee is planned as part of the Master Plan for Health 2021–2031 and the second five-year Strategic Human Resources for Health Plan for 2023–2027.

The strengths, weaknesses, opportunities, and threats (SWOT) analysis of the HRHIS revealed strengths such as the availability of staff for daily operations, a network of HRH managers, and the accessibility of the system from any location, along with projects aimed at extending its functionality to the private sector. However, there are also significant weaknesses, including the lack of operability of certain features (for example, HRH specialty data are not available), management of the database by a private provider, insufficient technological infrastructure, Internet connectivity issues, and the absence of a budget for its maintenance. Additionally, the digitization of personnel files, the expansion of the IT technician workforce at departmental focal points, improvements in information and communication technology infrastructure, and interoperability with other national systems remain pending.

One of the biggest challenges for HRH planning, based on the available data, is the age pyramid of health workers, with a large proportion set to reach retirement age by 2030. This poses serious problems due to the low recruitment caused by budgetary constraints.

Module 3. Overview of the national health workforce accounts

In this module, the conceptual and operational framework of the NHWA was presented: definitions, concepts, and measurement methodology. Below is a summary of the exercise conducted in two phases.

National health workforce accounts, mapping of data sources and actors, and measurement methodology

In line with Mr. Boniol's presentation, Teena Kunjumen from WHO emphasized the role of the NHWA as a system for countries to progressively improve the availability, quality, and use of data, therefore helping to advance universal health coverage. Implementation of the NHWA is primarily based on the specific needs of countries, data diversification, strengthening governance, active participation of focal points and national information systems, mutual collaboration, promoting the use of data for decision-making, increasing visibility of HRH, and resource mobilization. The data are collected on a common open-source data platform.

Based on the cumulative experience since the first year of implementation (2019), a second version of the NHWA indicators was developed, reducing its number of modules from eight to four (Density, Education, Finance, and Policies).

A total of 62 indicators were distributed across these modules. The focal points of each country are responsible for data input, although other users with different roles, such as managers, consultants, and other officials, can also be incorporated. These users have viewing and printing access to the information but are not authorized to edit the data.

Next, the speaker delved into the methodological details for data entry into the NHWA platform, focusing on the categories that present the greatest conceptualization challenges:

- Total number of health workers and the level of activity they perform;
- Ownership of facilities: public, private, and other sectors;
- Types of facilities: hospitals, community centers, clinics, etc.;
- Migration of health personnel: place of birth and place of training; and
- Training of health personnel.

Finally, as a precursor to the practical exercise, visual examples were shared on proper data entry into the NHWA platform, using its Health Personnel Education module as a reference.

Key definitions of indicators were also described, such as the proportion of health workers, the proportion of workers by age group, and the proportion of workers born abroad.

Practical exercise: Critical issues

Format: group work by country.

The first part of this practical activity consisted of group work focused on the most complex categories:

- Level of activity in the NHWA;
- Type of facility and ownership;
- Place of birth and place of training; and
- Education.

Each group analyzed one of these topics, identifying particularities and challenges, and proposing solutions (Image 3).

Subsequently, an individual exercise was conducted, where participants entered data in a test environment of the NHWA platform, using fictitious data on the prioritized indicators.

IMAGE 3 Participants in the regional workshop “Development of human resources for health information systems in the Americas” during the group practical exercise, Panama, 2023



© PAHO

Module 4. Methodology for occupation mapping in the development of human resources for health information systems

One of the opportunities offered by the NHWA platform is the standardization of information, in alignment with international standards. The ILO's International Standard Classification of Occupations 2008 (ISCO-08) provides a list and description of occupations across all sectors of activity.

This workshop module focused on occupations in the field of HRH. Experiences from different countries were presented, and the proposed methodology for occupation mapping (adapted to national contexts) was discussed. Chile and Costa Rica shared their experiences, followed by an analysis of occupations in the Region of the Americas and a practical activity for participants.

Chile: Mapping occupations in the health system

The representative of Chile, Claudia Godoy, began her presentation by providing context on the ministerial structure and the Chilean healthcare system, which utilizes a mixed model with participation by the public and private sectors.

She outlined the strategic approach to HRH planning, emphasizing the need for an information system that enables the anticipation of staffing requirements in a constantly evolving healthcare and organizational environment, thereby facilitating decision-making.

Regarding the occupational registration process in Chile's public healthcare system, Godoy emphasized that the primary definition of occupations is based on a solid legal and regulatory framework. Workers and groups of individual providers are registered with the Superintendence of Health; this forms the basis of the HRHIS, which covers healthcare workers nationwide. Chile has an occupational classification document adapted from ISCO-08, tailored to the country's specific context.

Among the main challenges mentioned were interoperability, the standardization of occupation names and codes, and the importance of establishing a registration process with designated personnel responsible for its maintenance and validation.

Costa Rica: Mapping occupations in the health system

Gloriana Mora Cascante, Head of the Human Resources for Health Unit at the Ministry of Health, provided an overview of Costa Rica's National Human Resources for Health Information System (SINARHUS). The system is characterized by the use of catalogs that standardize variable parameters, facilitating the unification of occupational nomenclatures. It classifies occupations into categories and subcategories based on professional training levels, as well as occupational groups that describe specific qualifications or specializations of personnel. The coding of occupations is based on ISCO-08, adapted to Costa Rica's context. The adaptation process of SINARHUS is dynamic and involves ongoing interaction between the Ministry of Health and different regions of the country. A key aspect is the prioritization of specific occupations based on each region's epidemiological and demographic profile. Currently, the system continues to work toward creating a more standardized nomenclature, harmonizing codes to enhance interoperability, incorporating new occupations, keeping the platform updated in real time, and automating processes.

Jamaica: Mapping occupations in the health system

Stephanie Bernard-Stephens and Nastasia Sancko, representatives of Jamaica, presented the 2015 Jamaica Occupational Classification System, which organizes similar occupations – derived from national censuses – into a structured list of categories. The main occupational groups include managers; technical and associate professionals; administrative support workers; service and sales workers; agricultural, forestry, and fishery workers; craft and related trades workers; plant and machine operators; assemblers; and the armed forces.

This classification is used as a reference by the Ministry of Health and Wellness, as well as other ministries, agencies, and departments. In Jamaica's Health Care Information and Visualization System (JaHRVis), a person's occupation must be selected from a drop-down list, with no free-text option.

Jamaica conducted an equivalency exercise between its national occupational classification and ISCO-08, resulting in a pyramidal structure with five distinct levels, each assigned a specific numeric code. The highest level is the major group (10), followed by the sub-major group (code 41), the minor group (129), the unit group (442), and finally, the specific occupations group (1071).

The current classification is highly specific and includes thousands of options, which sometimes leads to the field not being completed in JaHRVis if it is not mandatory.

Although there is no national prioritization of occupations, there is one at the ministerial level. For example, the reform of the primary health care system prioritizes the training of healthcare workers in areas such as family health, public health, and emergency medicine. Scholarship programs and specialized training in oncology and sub-specialties, as well as in medical physics, have also been implemented.

The next steps include creating a facilitating environment focused on HRH, which is a key component for achieving the reform of the primary health care system. The Ministry of Health and Wellness is developing a human resources for health plan, and progress is being made toward an electronic health information system.

Suriname: Mapping occupations in the health system

In 2021, Suriname had a population of 612 985. Its official language is Dutch, but a variety of languages and dialects are spoken due to the country's ethnic diversity, which includes Indigenous peoples, Surinamese Africans, Caucasians, Lebanese, Portuguese, Chinese, Javanese, Indians, and people of mixed ethnicities. Half of the population lives in coastal areas, while different tribes, primarily Indigenous and Maroons, inhabit the interior.

Muersia Williams, Senior Policy Officer at the Ministry of Health, reported that the occupational classification system in Suriname is based on functional analysis and the Suriname Government's information system (called FISO, in Dutch), which includes job descriptions and a salary structure. Since 2010, FISO has been applied to all government and semi-government occupations, and in some cases, it is also used as a guide in the private sector. A new structure is currently being developed for the healthcare sector.

Regarding the use of ISCO-08, Suriname had recognized and used the ISCO-88 version through the General Bureau of Statistics of Suriname (ABS, in Dutch). In 2017, the ABS developed a Safe Route for Socioeconomic Integration, although its application in the health sector is still unclear.

The process of comparison and alignment between ISCO-08 and Suriname's national system has recently begun. Current challenges include the incomplete registration of all health sector occupations at the Ministry of Health, the registration of dual or multiple specialties, and the incorporation of new occupations, some of which are hybrid.

To alleviate pressure on existing occupations, institutions are developing new positions, such as the training of high school students as surgical assistant technicians due to the shortage of nurses in operating rooms.

Suriname also faces shortages in specialized occupations, such as neonatal nurses for intensive care units and medical professionals specializing in emergency medicine, cardiothoracic surgery, neurosurgery, and geriatrics. The aging population increases the demand for medical professionals specializing in geriatrics, while the constant brain drain of paramedics leaving the country immediately after graduating further complicates the situation.

Mapping of occupations at the regional level

One of the challenges in the process of mapping occupations in the Region of the Americas is coordinating and harmonizing the wide variety of occupational classifications used across the Region. Mario Dal Poz, Director of the WHO Collaborating Center at the Institute of Social Medicine of the State University of Rio de Janeiro, presented an analysis of health occupations in the Region based on sources such as government websites and documents on professional occupations, comparing them with the NHA and ISCO-08. The review of websites showed that over 95% of countries have an occupational classification within their ministries of health, labor, education, statistics, and government. However, only 29 of the 35 countries consulted have a formal document defining professional occupations.

To advance the regional occupational mapping, the speaker proposed that countries import their various occupational classifications into a common platform. As a first step, he invited countries to identify their occupations using ISCO-08 as a reference and to create a list that includes both equivalent and non-equivalent occupations.

Next, the occupations with their local designations were input into a dynamic spreadsheet for each country, grouped according to the WHO NHWA coding and ISCO-08. The next step involved analyzing the documented occupations to standardize them, eliminate duplicates, and address those without an equivalent in the classification. Once standardization and normalization were completed, a comparison was made between the NHWA and each country's occupations to develop occupational profiles that could be exported as reports.

Practical exercise: Occupation mapping

This activity was carried out in small groups (Image 4). The first part involved critically analyzing and discussing the methodology and data sources used so far for the mapping of occupations, as well as the results obtained. Then, each country was asked to identify, describe, and provide examples of professions corresponding to occupational categories identified as priorities for primary health care (PHC), such as community health workers, dental assistants and therapists, dentists, pharmacists, psychologists, laboratory technicians, and nutritionists, or any others deemed relevant by each country. Finally, the results were recorded in a standardized model within a spreadsheet, which was made available in the PAHO Virtual Campus Community of Practice.

IMAGE 4 Group activity during the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023



© PAHO

Module 5. Methodology for mapping of actors in the development of human resources for health information systems

The importance of HRHIS governance was emphasized during the opening presentations of the session. A key step in building a governance framework is identifying and characterizing the actors who can contribute information to the HRHIS.

In this module, countries were invited to share their experiences, followed by a regional presentation and a final practical exercise.

Belize: Mapping actors in the human resources for health system

Lizett Bell, Deputy Director of Health Services and Chief Nurse at the Ministry of Health and Wellness of Belize, began her presentation by highlighting that although her country is recognized as part of the Caribbean, it is also considered part of Central America since it is located on the continent. It has a population of approximately half a million people, 56% of whom live in rural communities.

Belize has a single national university that offers training for nursing and health personnel, but it does not have a medical school. Medical professionals are trained abroad, primarily in Mexico, Guatemala, and other countries in Central America, with support from Cuba and Taiwan for specialized training. Training in nursing fields is also conducted at the University of the West Indies and the University of Technology in Jamaica.

Key actors in Belize's HRH system include professional councils and boards, the academic sector, other ministries in addition to the Ministry of Health, the private sector, and statutory bodies. The professional councils governed by law include the Nurses and Midwives Council, the Medical and Dental Council, and the Pharmacy Board, which is currently undergoing updates to mandate registration and continuing education. Health technicians are registered with the Ministry of Health through the Licensing and Accreditation Unit.

The University of Belize and its faculties of health sciences are part of the academic component. The health system is funded by the Ministry of Finance, while the Ministry of Education collaborates by providing scholarships and educational opportunities for healthcare workers. The private sector, in turn, includes some medium and small healthcare facilities. The national referral hospital is the Karl Heusner Memorial Hospital, a state-run entity.

For policy analysis and planning, the Ministry of Health collects HRH data through various mechanisms. However, one of the main challenges is the lack of integration of the private sector and the absence of a legal framework for collecting data on healthcare personnel in this sector.

In 2022, Belize digitized its human resources management system through a web platform. A module was launched for the public health system, with plans to include the Karl Heusner hospital in 2024. This tool allows for the transfer of data from Excel and generates various reports and tables for analysis.

Keeping the system continuously updated is challenging due to the frequent movement of healthcare staff, new hires, and the ongoing reliance on the paper-based registration system at the Ministry of Public Services. Maintaining the system's accuracy requires commitment at the local level, but the high turnover of administrative staff responsible for it necessitates continuous training.

El Salvador: National health workforce accounts for 2022

Paola Alejandra Cea Tenorio, a technical associate at the Directorate of Human Resources, outlined the institutional actors that make up the Integrated National Health System, led by the Ministry of Health. The system includes nine entities: the Ministry of Health itself; the Salvadoran Institute of Social Security; the Solidarity Health Fund; the Military Health Command; the Salvadoran Institute of Teacher Welfare; the Salvadoran Institute of Comprehensive Rehabilitation; the National Directorate of Medicines; the Ministry of Education, Science, and Technology; and a representative from private-sector hospitals.

For the data collection process of Module 1 of the NHPA, the Ministry of Health participated along with 31 national hospitals, five regional health offices, and representation from the central level, in addition to the institutions in charge of social security and military health. Records from the Higher Council of Public Health were used to identify all duly registered health personnel. Partial information was obtained on the main health professions in the country, including medical professionals, epidemiologists, psychologists, physiotherapists, nurses, dentists, pharmacists, and nutritionists, as well as health promoters and social workers. Finally, the representative from El Salvador outlined the challenges faced by the Ministry in developing its HRHIS, including: overcoming the lack of personnel for data analysis, establishing partnerships and regulatory mechanisms to share information with other government actors, incorporating the HRH module into the HRHIS, improving accessibility for all actors to upload data into the system, and conducting a national census of health personnel.

Paraguay: Integrated national human resources for health information system: Features, processes, and progress

Cristina Caballero García, Director of Paraguay's National Strategic Directorate of Human Resources for Health, began her presentation by describing the features, mission, and processes of the institution she leads, emphasizing its role as a governing and regulatory body in the field of HRH. She highlighted the importance of Paraguay's National HRH Policy 2020–2030 and its implementation plan, which is structured around three key strategic lines: i) strengthening and consolidating the governance and strategic planning of HRH policies; ii) implementing the Integrated National HRHIS; and iii) developing supportive environments and capacities of HRH to achieve universal health coverage and access, with a focus on primary health care (PHC) and integrated health service networks. The speaker described the regulatory and legal advances that have strengthened HRHIS governance, such as the creation of a national technical committee, data collection strategies, and the methods and technological tools for the Integrated National HRHIS, as well as the development of educational capacities through a national training and development plan. Regarding communication strategy, she emphasized the importance of national publications on the characterization of health professionals, multiple job-holding, and HRH baseline data. Finally, she provided an analysis of current regulatory strategies and the creation of catalogs of medical, nursing, and dental professionals.

Regional mapping of actors

Isabel Duré, a member of HSS/HR, provided an overview of HRHIS, the available data sources, and the proposed strategy for countries' mapping of actors and technical cooperation. She highlighted the HRHIS structuring model, which is based on administrative records of professionals, health facilities, and system users. This model is progressively gaining prominence compared to the traditional model, which relied on censuses and surveys. Among the available data sources, she mentioned health facilities, registrations, certifications, degrees, legalizations, migrant records, and employment contracts, as well as tax-related information and identity validations.

Regarding mapping of actors, she highlighted that this methodology facilitates the assessment of the interest, influence, impact on policy, and resources of key actors for the development of HRHIS. The objective is to bring these actors together to develop a common road map toward an integrated information system.

The proposed methodology consists of the following four steps: i) identification of actors; ii) regulatory analysis; iii) interviews and political dialogues; and iv) technical analysis. The main outcome of this methodology is the establishment of alliances and agreements to strengthen HRH and their respective information systems.

Practical exercise: Mapping of actors

During this activity, participants were invited to discuss and share insights in groups on one of three key topics related to information systems, focusing on how to address them through technical cooperation in human resources information systems.

The conclusions were recorded in the VCPH forum.

The overall results are presented in Part four: Results and conclusions.

Module 6. CARICOM's human resources for health caribbean commission

The third day of the workshop was focused on collaborative work with all the countries of the Region.

The countries of the Caribbean Community (CARICOM) have a human resources commission that works in an integrated manner to strengthen HRH through strategic policies and actions.

The opening of this session highlighted the progress made by this commission.

Human resources for health caribbean commission

Helen Royer, Director of Human Development at the Department of Human and Social Development of the CARICOM Secretariat, presented CARICOM's Human Resources for Health Caribbean Commission, which is made up of 20 countries. Of these, 15 are Member States and 5 are Associate States, collectively representing a population of approximately 16 million people.

The speaker addressed the challenges faced by the subregion, such as the shortage of essential personnel, including medical and nursing professionals, as well as midwifery staff. The retention of health personnel is a critical issue, as many emigrate in search of better working and financial conditions. She also pointed out the lack of specialists in key areas such as neurology, cardiology, and oncology, as well as the increasing demand for professionals in mental health and geriatrics due to the aging population.

The Commission has set out to improve governance and leadership in the health sector, as the capacity to regulate and ensure the quality of both public and private health systems is limited. Priorities include enhancing information management, strengthening national and regional capacities, and using strategic information to support decision-making.

The Commission was established in November 2020 in response to the COVID-19 pandemic and other health emergencies. It was initially created as the Health Human Resources Action Task Force (HRH-ATF) and later formalized as a standing committee called the Human Resources for Health Caribbean Commission (HRH-CC).

The Commission is also working on harmonizing regulations for health professionals and providing training in various Member States. In addition, online training programs have been developed, and work is underway to review nursing curricula in collaboration with PAHO.

One significant challenge is the sustainability of programs, many of which are discontinued once donor funding ends, highlighting the need to develop sustainability plans. The importance of research and the application of its findings to address the specific needs of Member States was emphasized.

Finally, the speaker highlighted that a potential restructuring of the commissions and working groups is being reviewed to better respond to the needs of Member States.

From the action task force to the human resources for health caribbean commission: An overview

Benjamín Puertas presented the evolution of the Human Resources for Health Caribbean Commission and its work on key issues related to the management and development of health teams in the region, highlighting the transition from the HRH-ATF to a standing committee as a significant milestone.

The Commission has collaborated with various institutions, such as the Caribbean Public Health Association (CARPHA), the University of the West Indies, and the University of Guyana, in the development of policies aimed at strengthening human resources, with a particular focus on pandemic response and addressing vaccine hesitancy among healthcare workers.

Additionally, he emphasized the importance the Commission places on HRH planning and forecasting, which involves the use of the NHWA. With the support of PAHO and WHO, key indicators were established to monitor and improve the health workforce in the Region.

Education and training were also key aspects, with the creation of online training programs and the setup of simulation labs in several member countries. The Commission's progress was driven by active participation from the countries and collaboration with national and international academic institutions. However, challenges were also encountered, such as limited financial resources and unequal participation among the countries.

A key initiative was the launch of the Americas Health Corps, in collaboration with the United States Government and PAHO, aimed at training 500 000 healthcare workers in Latin America and the Caribbean over a five-year period. This program, launched with initial funding, seeks to strengthen national health systems and address critical issues such as the migration of healthcare workers.

The Commission focuses on strengthening HRH through strategic policies and actions, international collaboration, and educational capacity-building, in an effort to improve the quality and accessibility of healthcare services in the region.

Basic human resources for health indicators in the Caribbean

The Human Resources for Health Caribbean Commission is tasked with monitoring and advising on priority issues related to HRH, including the migration of healthcare workers. Additionally, it seeks to align national HRH policies with regional strategies.

Fiona Harris, International PAHO Consultant for the Caribbean, explained the set of basic and supplementary indicators initially agreed upon by the HRH-ATF, based on the WHO NHWA indicators. A list of 30 indicators was established and is implemented through the NHWA system.

The indicators, selected based on the feasibility of data collection, the number of actors involved, and their relevance to the region's needs, are grouped into three levels:

- Level 1: consists of 13 basic indicators that all countries must meet.
- Level 2: includes 13 supplementary indicators for countries that have already met Level 1 requirements.
- Level 3: contains 3 additional indicators for countries that have completed the previous levels.

Level 1 covers indicators related to the active health workforce, education and training, employment characteristics, governance, and HRHIS. Level 2 includes indicators related to workforce and flow of workers, regulation of training, and health labor market flows. Level 3 focuses on voluntary and involuntary exits from the health labor market, as well as vacancy rates.

The goal of this agreement on indicators is to provide Caribbean countries with essential data to make informed decisions regarding HRH. However, countries face challenges such as the lack of comprehensive HRH information systems, inefficient and fragmented data collection, and limited capacity in HRH within ministries. Furthermore, few countries have national HRH plans and strategies that prioritize this issue.

The speaker highlighted the collaboration between the Commission and statisticians from several countries, which resulted in an action plan to implement HRH indicators. For its implementation, essential priority areas were identified, such as governance, capacity development, health information management systems, and the strategic use of data.

The use of tools and resources provided by PAHO, WHO, and the VCPH can help strengthen capacities and improve the HRH situation in the Caribbean.

Module 7. Human resources for health information system maturity assessment

The various divisions of PAHO responsible for HRH indicators work interprogrammatically to optimize data collection and improve the quality of published data. In collaboration with the Department of Evidence and Intelligence for Action in Health, information is coordinated on various health professions, such as medicine, nursing, dentistry, pharmacy, and technical areas, leveraging best practices in the development of information systems.

This approach lays the foundation for the implementation of the HRH Information Systems Maturity Assessment Tool, which is introduced in this module.

Information systems for health: Conceptual framework, information systems for health maturity model, and development of human resources for health information systems in the Americas

Andrea Gerger, Advisor in the PAHO Department of Evidence and Intelligence for Action in Health, presented the conceptual framework for Information Systems for Health (IS4H), which is championed by PAHO, along with its maturity model (IS4H-MM). The IS4H initiative integrates the Plan of Action for Strengthening Information Systems for Health 2019–2023 approved by PAHO Member States. It was designed to improve universal access to health and universal health coverage in the Region by strengthening interconnected and interoperable information systems. Its main purpose is to generate quality information that enhances medical, management, and health policy decision-making. In line with this, it is structured around four strategic objectives: i) data and technology management; ii) management and governance; iii) innovation and performance; and iv) knowledge management and exchange. Each objective

involves key components to ensure the effectiveness of IS4H. Additionally, PAHO provides policies, evaluation tools, documentation, guides, educational materials, and a community of practice to collaborate with countries in the development of their information systems for health. Speaking about the IS4H-MM tool, she noted that it is a maturity analysis model that conducts assessments at the national level through an IT system based on the four strategic objectives and their components. The model has five levels of maturity, with Knowledge Management and Exchange being the most advanced segment of the model in the Region. Finally, the speaker highlighted the usefulness of the model's results as a tool for setting strategic priorities and workplans within the IS4H initiative; as input and an approach for process improvement across the various IS4H areas; as a basis for continuous monitoring of IS4H capacity and process improvement; as critical input for the digital transformation of the health sector; and as a benchmark for IS4H maturity, enabling other national health authorities in Latin America and the Caribbean to adopt best practices (Image 5).

IMAGE 5 Presentation of the IS4H Maturity Model at the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023



© PAHO

Building on the initiative of the Department of Evidence and Intelligence for Action in Health, the Human Resources for Health Unit has developed a proposed tool to assess the maturity of information systems. As part of the workshop, representatives from the Bahamas, Panama, and Uruguay were invited to present the maturity status of their HRHIS. This was followed by the presentation and validation of the tool to assess HRHIS maturity at the regional level.

Bahamas: National committee on human resources and health workers

Paul Haven, Vice President of Human Resources at Doctors Hospital Nassau, presented on the National Committee on Human Resources and Health Workers in the Bahamas. He discussed the four strategic objectives that the committee has defined: i) strengthening and consolidating governance and leadership in HRH, by establishing strong cross-sectoral relationships; ii) optimizing the performance of health personnel; iii) implementing a health workforce information system; and iv) facilitating the development and funding of a national HRH policy and action plan. In this context, he noted that the IS4H initiative has helped connect the HRH component with other essential areas, such as service delivery, governance, health technologies, infrastructure, and financing. For the IS4H maturity evaluation process, the Bahamas applied PAHO's IS4H-MM model, covering the following domains: i) data and technology management; ii) management and governance; iii) innovation and performance, and iv) knowledge management and exchange. The country received a score of between 2 and 3 on a scale with a maximum level of 5. Key recommendations from the evaluation process included: strengthening best practices, domains, and components of IS4H; updating the national legal and regulatory framework; establishing standardized operating procedures; enhancing national coordination and planning through collaborative governance and priority setting; improving communication channels; and identifying needs and appropriately allocating financial resources.

Panama: The human resources for health information system: A key pillar in health system development and policy-building

Dianne Torres Quintero, Human Resources Planner at the Ministry of Health, presented the background of Panama's HRHIS. She explained that initially, data were managed manually using Excel spreadsheets, with non-standardized variables that created numerous problems. In response to this, a goal was set to digitalize and modernize the information on the health sector workforce to improve decision-making in administrative, human resources management, and health services at the national, subnational, and local levels. Currently, the country has a technological platform for managing the HRHIS, along with a dashboard that displays key reports on the indicators.

The indicators for the HRHIS were extracted from the *Handbook for measurement and monitoring indicators of the regional goals for human resources for health*¹ and the *National health policy and strategic guidelines 2016-2025*² of the Republic of Panama. These indicators focus on aspects such as the density of health professionals per 10 000 population (medical and nursing professionals, and midwifery staff); the ratio of medical professionals per inhabitant working in primary health care (generalist medical professionals/specialist medical professionals and nursing professionals/medical professionals); the specific density of different types of professionals per 1000 population; or the distribution of professionals and other health workers by competencies. The IS4H Maturity Model was applied to assess the maturity of the HRHIS, yielding positive results in data management, governance, and knowledge exchange.

Uruguay: Maturity assessment of the information system

María Lucía Aicardi, a member of the Ministry of Public Health, emphasized the importance of HRH in the healthcare process, as this represents approximately 70% of healthcare spending and 7% of the total workforce in Uruguay. She highlighted the Ministry of Public Health's goal of consolidating its regulatory role through the HRHIS, a priority for Uruguay in its efforts to strengthen this area.

¹ Pan American Health Organization. Handbook for measurement and monitoring indicators of the regional goals for human resources for health: a shared commitment. Washington, D.C.: PAHO; 2011. Available from: <https://iris.paho.org/handle/10665.2/31098>.

² Ministry of Health of the Republic of Panama. Política nacional de salud y lineamientos estratégicos 2016-2025. Panama: Ministry of Health; 2016. Available from: <https://faolex.fao.org/docs/pdf/pan174954.pdf>.

In this regard, the speaker explained that the HRHIS in her country includes a system for the licensing and registration of healthcare professionals, which regulates the registration of qualifications as a means of oversight. The system encompasses the administrative records of personnel working for the Ministry of Public Health, decentralized emergency entities, highly specialized services, and other public or private centers that provide information through their databases. She also highlighted that the Ministry's administrative records cover more than 72% of healthcare workers.

Regarding the maturity of the HRHIS, the speaker described the key assessment categories: i) data management; ii) information technologies; and iii) stewardship and governance. The levels of progress vary by the areas assessed: workforce and flow (11 out of a maximum level of 12); education (5 out of a maximum of 13); spending and financing (2 out of a maximum of 5); and working conditions, governance, and leadership (23 out of a maximum of 28). In conclusion, she emphasized that Uruguay has consolidated strong stewardship of information systems, with a focus on the development of enforcement mechanisms for generating sector data.

Although significant progress has been made, a unified information system has not yet been achieved, meaning there is still progress to be made toward a joint HRH planning framework.

Presentation of the proposed maturity model for the human resources for health information system at the regional level

Michele Guillou, an international consultant for PAHO's HRH Unit, provided an analysis of the concept of "maturity model" and defined full maturity as a state of full development in which an organization, process, or project has achieved the maximum performance level. The HRHIS maturity model presented draws on PAHO's IS4H-MM as a reference. However, unlike the IS4H-MM, which consists of five progressive levels, the HRHIS maturity model is made up of four levels.

The design of this model was based on the following attributes: self-assessment, alignment with the NHWA, efficiency in terms of time and effectiveness, its potential to strengthen HRHIS plans, and progressive utilization. The model was structured around three strategic areas: i) data management and information technologies (DMIT); ii) management and governance (MAGO); and iii) knowledge and decision-making (KDM). For each of these areas, four levels of maturity (from 1 to 4) were established. Spreadsheets were used to develop the tool for its implementation.

Finally, the speaker highlighted some key suggestions for assessing the maturity level of each HRHIS in the countries of the Region and led an activity to validate the model with the workshop participants.

Practical exercise: Maturity model for human resources for health information systems

Working groups were formed, and participants were guided to discuss and share opinions on:

1. The structure of the tool.
2. Specific questions related to each of the three strategic areas of the tool.

The tool, created in spreadsheet format, was used for a maturity assessment exercise. It includes a home tab with instructions and general information, a tab for each strategic area with its respective four levels, and a final tab with a glossary of terms and concepts (Image 6). A document with validation guidelines was also provided to better guide the exercise.

IMAGE 6 Screenshot of the tool designed to assess the maturity of human resources for health information systems, Panama, 2023

Measurement and Maturity Analysis of Human Resources for Health Information Systems - VERSION 13/09/2023

Strategic Areas

Note 1: Instrument adapted from the Information Systems for Health Maturity Tool, IS4H, developed by PAHO and WHO

Note 2: Instrument based on the National Health Personnel Accounts.

Instructions for using the HRHIS maturity analysis tool:

1. Save this document with the name of your country or institution, and the date.
2. Provide the information requested below regarding the organization's profile.
3. Answer each question of the three categories (strategic areas) by selecting the number that best describes the current situation of your organization under the column "Level".

Provide the following information about your organization

Name(s)

Date

Location

Organization

Additional information

Navigation: Inicio | GDTI | GEGO | COTD | Glosario | +

Source: Pan American Health Organization. Virtual Campus for Public Health [website]. Washington, D.C.: PAHO. Available from: <https://campus.paho.org/en>.

The groups found the tool appropriate for assessing HRHIS maturity and provided suggestions for its improvement within the community of practice for strengthening HRHIS, which are available on the VCPH (Image 7).

IMAGE 7 Participants in the regional workshop “Development of human resources for health information systems in the Americas” during group work, Panama, 2023



© PAHO

PART THREE

Challenges and priorities of countries in the region for the development and strengthening of human resources for health information systems

Through the VCPH Community of Practice, a survey was conducted on challenges and priorities in the development and strengthening of HRHIS in each country, with the aim of guiding technical cooperation (Tables 2 and 3). The results reflect needs and requirements covering various areas of HRH management.

Challenges and priorities for Latin American countries

Bolivia (Plurinational State of)

1. Updating the human resources policy and developing an action plan for HRH, with the human resources system as one of the key strategies. To create this national document, it is essential to take into account the mapping of actors and information systems carried out in 2021 by consultant Juan Lucas Pereyra.
2. Integrating information systems from the social security subsectors in the short term, including both public and private sectors.
3. Implementing a catalog to optimize the country's professional registration system, with the involvement of relevant sectors led by a national human resources committee.

Chile

1. From the perspective of technical cooperation, and considering the local context, the main challenges are related to implementing interoperability among different systems, which involves standardizing nomenclatures and codes.
2. In addition, receiving support to increase the visibility of these issues in the country is crucial, as backing from authorities is needed to advance intersectoral coordination and make key decisions.

Costa Rica

1. Implementing a strategic communication plan with a focus on change management, aimed at the actors responsible for generating information to monitor the policy.
2. Receiving guidance in developing intermediate indicators and implementing data collection, to achieve proper measurement of the established indicators, with an emphasis on validation and quality.

3. Currently, these are the two main areas of collaboration identified, as the country is in the process of developing and approving its policy. However, after this stage, support may be needed in other areas to strengthen both the policy and the HRHIS.
4. Additionally, there is an identified need for technical support to assess the maturity of the existing information system in the country.

Cuba

1. Support for the continuous training of officials and managers responsible for human resource management at various levels of the national health system.
2. Working toward the unification of information systems.
3. Strengthening the computerization of information systems management.

Dominican Republic

1. Establishing the governance framework of the HRHIS, specifying roles, data collection frequency, monitoring processes, information security, interoperability, data protection, and cybersecurity; and ensuring that the system itself integrates primary sources.
2. Launching a strategic communication campaign to increase visibility of the topic both within and outside the Ministry of Health.
3. Technical support for the development and implementation of the HRHIS, including georeferencing, self-registration, validation processes, external consultations, cybersecurity measures, and integration of migration-related topics.
4. Receiving support for the development of the first HRH census.

Ecuador

1. The National Human Resources Policy is currently under development, and it is essential to address the critical issues that require support:
 - Human resource attrition;
 - Inefficient use of personnel;
 - Low productivity; and
 - A dichotomy between the public and private sectors.

Guatemala

1. Developing a common catalog that includes all institutions that can provide information, which would improve the quality of data.
2. Creating a national HRH register.
3. Establishing legal frameworks that mandate the submission of data to address the frequent lack of necessary information caused by actors' reluctance to provide it.
4. Implementing an approved human resources policy.
5. Strengthening the Ministry of Health's leadership of other actors, such as social security, universities, professional associations, and private services.

Panama

1. Technical support to advance the implementation of the HRHIS at the Ministry of Health and technical advice to foster collaboration with other actors in the HRH sector, such as the Social Security Fund, employers' organizations, and training institutions (public and private universities).
2. Technical advice for the strategic development of Panama's National Human Resources Plan.
3. Technical support to update the status of human resources (diagnosis, gaps, and workforce analysis).

Paraguay

1. Continuing collaboration efforts aimed at fulfilling Strategy 3 of the 2020–2030 Paraguayan National Human Resources for Health Policy;³ particularly with the implementation of the national strategic human resources in health information system.
2. Planning, integrating, and standardizing a strategic HRH information system that provides data for management, planning, and decision-making, with an approach that encompasses the entire national health system, in order to advance the following topics:
 - Creation of an integrated register, receiving input from multiple primary sources and enabling the seamless integration of data across the entire national health system through Paraguay's Resources for Health Observatory; and
 - Standardization of electronic records among institutions.
3. Ensuring interoperability among information systems to exchange data.

Peru

1. Standardizing the country's catalog of occupations.
2. Advancing the approval of a regulatory framework (currently under review by the national health authority) that mandates health personnel registration by institutions.
3. Facilitating the interoperability of the national health personnel register with the immigration office's data management system.

Uruguay

1. Encouraging broader dissemination of the NHWA, its results, and its benefits to increase adoption by health authorities and improve uptake of information systems.
2. Integrating additional health sector entities into information systems, such as private aesthetic clinics, ophthalmology and dental practices, laboratories, long-term care facilities, and pharmacies.
3. Building partnerships to enhance interoperability with other databases, such as social security, death records, and the Ministry of Education.

TABLE 2 Summary of technical cooperation needs and requirements in human resources for health in Latin American countries, in order of priority, 2023

Requirement/country	BOL	CHL	CRI	CUB	DOM	ECU	GTM	PAN	PAR	PER	URY	Total
HRH policies and plans												6
Prepare HRH policy							4					1
Strengthen/update HRH policy	1		3			1 ^a						3
Develop HSH action plan	1											1
Develop national HRH plan								2				1

³ Ministry of Public Health and Social Welfare of Paraguay, Pan American Health Organization. Política nacional de recursos humanos en salud del Paraguay 2020-2030. Asunción: Ministry of Public Health and Social Welfare; 2021. Available from: <https://www.mspbs.gov.py/dependencias/portal/adjunto/43b0d2-POLITICANACIONALDERECURSOSHUMANOSSENSALUDELPARAGUAY.pdf>.

TABLE 2 Summary of technical cooperation needs and requirements in human resources for health in Latin American countries, in order of priority, 2023 (*continued*)

Requirement/country	BOL	CHL	CRI	CUB	DOM	ECU	GTM	PAN	PAR	PER	URY	Total
Regulatory frameworks												2
Develop a legal/regulatory frameworks to mandate the submission of information							3			2		2
Governance												4
Strengthen Ministry of Health governance							5					1
Strategically communicate the importance and usefulness of human resources for health information systems (HRHIS)			1		2						1	3
Development/strengthening of HRHIS												3
Strengthen/implement HRHIS					3			1	1			3
Interoperability												7
Achieve interoperability and sectoral integration in information systems	2	1		2				1	3	3	3	7
Development of a catalog/registry												8
Develop a common catalog among sectors	3						1					2
Capture data from other sectors									2		2	2
Create a national HRH register					4		2					2
Standardize and catalog occupations		2								1		2
Operational												2
Develop intermediate indicators, implement data collection			2									1
Strengthen the digitalization of HRHIS management				3								1
Maintenance, use, and governance of HRHIS												3
Improve information system governance					1							1
Assess system maturity			4									1
Data utilization								3				1
General development of HRH												1
Ongoing and broad training on HRH topics				1								1

Note: BOL: Bolivia (Plurinational State of); CHL: Chile; CRI: Costa Rica; CUB: Cuba; DOM: Dominican Republic; ECU: Ecuador; GTM: Guatemala; HRH: human resources for health; PAN: Panama; PAR: Paraguay; PER: Peru; URY: Uruguay.

^a Includes subtopics such as HRH attrition, inefficient use of personnel, low productivity, or the dichotomy between the public and private sectors.

Source: PAHO.

Challenges and priorities for Caribbean countries and territories and Canada

Anguilla

1. Technical support from PAHO to assist the Ministry of Health in developing policies for the integrated HRH system, with the aim of strengthening data management.
2. On-site training provided by PAHO to help identify key personnel for collaboration in the development and maintenance of the system.
3. Development of a strategic plan to address data fragmentation and creation of a semi-automated, integrated system that regularly updates authorities with relevant, accurate, and timely information.

Antigua and Barbuda

1. Ensure access to specialized knowledge for system support and maintenance.
2. Engage an IT specialist.
3. Hire a specialist in health information systems and medical records.
4. User support.

Bahamas

1. Training key actors involved in data management and analysis.
2. Increasing interoperability across information systems throughout the healthcare system.
3. Ensuring the necessary infrastructure and hardware to meet the specified objectives.

Barbados

1. Assistance in updating the country's HRH policy and action plan.
2. Assistance in implementing the recommendations derived from IS4H-MM, particularly in developing the 2019–2023 Action Plan for Strengthening Health Information Systems.
3. Evaluation of the HRHIS, with appropriate recommendations and an action plan.
4. Training of the Barbados Accreditation Council staff as soon as possible on the use of NHWA.
5. Strengthening the capacity to link HRH data with broader health information, such as population needs, service utilization, and patient outcomes.
6. Assistance in identifying an appropriate digital health information system to support the HRHIS.

Belize

1. Promoting the development of capabilities related to the NHWA with all relevant actors.
2. Creating infographics and profiles for the different categories of HRH.
3. Developing HRH profiles for the different levels of care.
4. Training government officials in the use of the NHWA.
5. Establishing a mechanism for conducting labor market analyses for healthcare workers.

British Virgin Islands

1. Technical assistance in collecting the necessary data to build the information systems.
2. Training for the focal point in the British Virgin Islands, as there is currently no training in place.

Canada

1. Microsimulation and projection in planning.

Dominica

1. Assistance in drafting and developing a human resources policy.
2. Capacity-building for training that facilitates participation and the use of the NHWA, and ensuring sustainable access to training nationally.
3. Facilitating technical exchanges focused on continuous learning and the development of HRH programs.

Grenada

1. Standardizing data collection instruments.
2. Training key actors in information management.

Guyana

1. Standardization of tools and methodologies for data collection.
2. Building capacities to engage focal points and key actors in the Region in data collection.
3. Technical assistance for establishing an HRH unit within the Ministry of Health.
4. Technical assistance for personnel management in the health sector.
5. Technical assistance for developing a workplan with academic institutions aimed at training human resources.
6. Technical assistance for restructuring health workforce training and the role of the Health Sciences Unit to establish a national institution dedicated to the development of health personnel.

Jamaica

1. Completion of the HRH framework and policy, along with its corresponding implementation plan.
2. Mapping of occupations and key actors in the health sector.
3. Training on data entry into the NHWA.
4. Development of a national HRH information system that includes cooperation from the private sector.
5. Ongoing training on the NHWA.

Montserrat

1. Training of key actors on HRHIS.
2. Assistance for the development of a national HRH policy.
3. Technical assistance to improve the functionality of information systems at the local level.

Saint Kitts and Nevis

1. Technical assistance for the development of the digital HRH system.
2. Technical support for updating the HRH action plan.
3. Technical assistance for training actors and relevant parties in the collection of HRH and NHWA indicator data.

Saint Lucia

1. Development of an HRHIS policy or strategy that includes the adoption of standardized job titles, codes, and job descriptions across all involved agencies.
2. Review of the proposed or recently implemented HRHIS to assess its capacity to provide the necessary indicator data and required analyses for decision-making (Is it feasible as is? Does it need customization, or is a completely new solution required?).
3. Review of the proposed or recently implemented HRHIS to determine whether it allows adequate access to data by actors and produces reliable statistics and indicators for decision-making.

Saint Vincent and the Grenadines

1. Review, updating, and expansion of the national human resources policy and action plan, including costs.
2. Development of an implementation plan for the HRHIS strategy.
3. Training of trainers for HRH and HRHIS focal points.
4. Support for integrating the HRHIS into the health information system.
5. Technical cooperation to continue HRHIS implementation.
6. Enhancing the capacity of the focal points.
7. Potential creation of a human resources unit.

Suriname

1. Assistance for the development of the HRH policy or strategy.
2. Support for the implementation of the HRHIS, ensuring alignment with the NHWA.
3. Capacity-building through training for the Ministry of Health and other actors.
4. Completion of the development and implementation of the HRHIS at the national level.
5. Training of Ministry of Health staff and actors on HRHIS in the country.
6. Development of the HRH action plan for the country.

Trinidad and Tobago

1. Standardization of data capture tools.
2. Training and awareness-raising for relevant actors.
3. Development of the HRH policy.

Turks and Caicos Islands

1. Implementation of long-term training and capacity-building programs to ensure that Turks and Caicos Islands has the necessary expertise to independently manage the functions of the HRHIS.
2. Training in quality assurance processes to ensure that the data collected and managed in the HRHIS meet high standards of accuracy and reliability.
3. Interoperability to ensure that the HRHIS is capable of exchanging data with other health information systems and databases, both nationally and potentially with regional or international systems.

TABLE 3 Summary of technical cooperation needs and requirements in human resources for health in the Caribbean and Canada, in order of priority, 2023

Priority/country	ATG	AIA	BHS	BRB	BLZ	CAN	DMA	GRD	GUY	JAM	MSR	LCA	VCT	KNA	SUR	TCA	TTO	VGB	Total
HRH policies and plans																			10
Development of the HRH policy		1					1			1	2				1		3		6
Development of an HRH action plan															3				1
Update of the HRH policy and/or action plan				1									1	2					3
HRH governance																			7
Creation of HRH Unit in the Ministry of Health									3				7						2
Creation of a national institution for the development of HRH									6										1
Increasing visibility of HRHIS and the NHWA					1						1				2		2		4
HRHIS development plan																			3
Development of the HRHIS strategic plan		3											2						2
Development of the IS4H action plan				2															1
HRHIS development																			6
HRHIS development		2								4			5	1	1,2				6
HRHIS sustainability																			8

TABLE 3 Summary of technical cooperation needs and requirements in human resources for health in the Caribbean and Canada, in order of priority, 2023 (*continued*)

Priority/country	ATG	AIA	BHS	BRB	BLZ	CAN	DMA	GRD	GUY	JAM	MSR	LCA	VCT	KNA	SUR	TCA	TTO	VGB	Total
HRHIS maintenance and support training	1		3	6							3					1			5
Identification of specialized knowledge for HRHIS support	2, 3, 4																		3
Data collection																			6
Establishment of standards for data collection								1	1								1		3
Capacity-building for data collection									2					3				1	3
Capacity-building																			11
NHWA platform training				4	4		2	2		3, 5			3, 6		2			2	10
Quality assurance training																2			1
Display/use of data																			6
Use of data for decision-making			1	5	2, 3							2, 3							6
Interoperability																			5
Interoperability			2									1				3			3
Integration of HRHIS into health information systems													4						1
Mapping of occupations and actors										2									1
Assessment																			1
HRHIS assessment				3															1

TABLE 3 Summary of technical cooperation needs and requirements in human resources for health in the Caribbean and Canada, in order of priority, 2023 (*continued*)

Priority/country	ATG	AIA	BHS	BRB	BLZ	CAN	DMA	GRD	GUY	JAM	MSR	LCA	VCT	KNA	SUR	TCA	TTO	VGB	Total
Planning																			2
Microsimulation and projection						1													1
HRH planning									4										1
Other HRH																			3
Labor market analysis					5														1
Development of an HRH training plan in conjunction with academia									5										1
General HRH training							3												1

Note: AIA: Anguilla; ATG: Antigua and Barbuda; BHS: Bahamas; BLZ: Belize; BRB: Barbados; CAN: Canada; DMA: Dominica; GRD: Grenada; GUY: Guyana; HRH: human resources for health; HRHIS: human resources for health information system; IS4H: Information Systems for Health; JAM: Jamaica; KNA: Saint Kitts and Nevis; LCA: Saint Lucia; MSR: Montserrat; NHWA: National Health Workforce Accounts; SUR: Suriname; TCA: Turks and Caicos Islands; TTO: Trinidad and Tobago; VCT: Saint Vincent and the Grenadines; VGB: British Virgin Islands.

Source: PAHO.

PART FOUR

Results and conclusions

A brief summary of the workshop

The workshop was conducted in two stages: the first with participants from Latin America and the second with participants from the Caribbean. All countries in the Region but two took part. In addition to the regional team, two officials from the WHO NHWA team were also present (Annex 3).

The workshop agenda was repeated on the first and second days for the Latin American and Caribbean countries, followed by a joint session with all participants (Annexes 1 and 2). Wednesday was a day of joint work for the entire Region. The agenda was structured into four modules focusing on different topics: information systems and the NHWA, occupation mapping, information system maturity, and mapping of actors. Each module included two country presentations, a conceptual presentation by experts, and a group activity. The NHWA Team from WHO, together with the regional team, provided training on the platform and assisted some participants in uploading their data during the workshop. Additionally, the VCPH team recorded the group discussions and shared the materials via the Moodle platform.

Results

Thanks to the effective organization and the structured modules, the workshop achieved the expected results. Additionally, it helped identify new challenges and priorities for technical cooperation. The following key contributions were made by the participating countries:

- Analysis of the regional and global status of HRHIS;
- Overview, exchange, and training of countries on the NHWA framework and methodology;
- Methodology for mapping health occupations, with countries committing to its implementation;
- Proposal of a methodology for mapping key actors in the development of HRHIS, presented and discussed with the countries;
- Validation of the methodology for assessing HRHIS maturity;
- Identification and documentation of the countries' challenges and priorities to strengthen HRHIS within the VCPH Community of Practice.

Conclusions

Significant differences were observed among countries in terms of their progress in the development and governance of their information systems; this highlights the need to tailor cooperation efforts to each country's specific circumstances.

During the workshop, country representatives examined the mapping of health occupations, sharing experiences and strategies to enhance the accuracy and completeness of records. The discussion aimed to develop a more

comprehensive and up-to-date understanding of the health workforce. Participants emphasized the importance of consistently categorizing occupations to improve the management of human resources for health.

The assessment of HRHIS maturity was a central topic of the workshop. Participants exchanged information about the current state of development of their systems, identifying challenges and opportunities to enhance their effectiveness. They also highlighted the importance of implementing the maturity assessment tool.

The need to invest in technology and training was also discussed to ensure that systems meet the growing demands of the health sector. Participants shared their progress in data collection and analysis, emphasizing the importance of closely monitoring the size and composition of the health workforce. Additionally, the discussion addressed the challenge of using these data for informed decision-making on health policies and strategic planning.

Regarding mapping of actors, participants explored strategies to identify the various actors involved in HRH management, including training institutions, regulatory bodies, unions, and other key allies. The focus was on strengthening governance through collaboration and coordination among actors, aiming for more efficient and effective health workforce management.

The countries had the opportunity to record their technical cooperation needs and priorities. All materials and suggestions are available through the VCPH.

The workshop proved to be valuable in promoting cooperation and knowledge exchange among the countries in the Region, and in building networks for improving HRH management.

Each of the topics discussed contributed to advancing the understanding and capacity of health systems, particularly in Latin America and the Caribbean.

Recommendations

- Continue providing direct support to countries facing the greatest challenges in their information systems, particularly in the Caribbean, but also in countries such as Honduras in Central America and the Plurinational State of Bolivia in South America.
- Promote the implementation of exercises on mapping of actors and the assessment of information system maturity in countries across the Region, while also encouraging the exchange of best practices.
- Develop and implement the methodology and tools for mapping of actors, HRHIS maturity assessment, and health occupation mapping to facilitate the development of HRHIS in countries across the Region.
- Continue strengthening data entry into the NHWA and advance health occupation mapping at the regional level, ensuring that each country has its own perspective while remaining aligned with the NHWA.
- Support countries in effectively communicating available information, whether through the publication of articles or other forms of dissemination.
- Conduct an in-depth analysis of the technical cooperation requests recorded by countries in the VCPH and apply the methodologies presented in the workshop, tailoring them to the specific conditions and needs of each context.

Annexes

Annex 1.

Regional workshop agenda: “Development of human resources for health information systems in the Americas” for Latin American countries, 11–13 September 2023

Day 1: 11 September

Time	Activity	Speaker/presenter/moderator
8:00–8:30	Participant registration	
8:30–9:20	Welcome and opening remarks	Benjamín Puertas, Pan American Health Organization (PAHO) Leah-Mari Richards, PAHO, Panama Dagoberto Vega, Ministry of Health, Panama
9:20–10:30	Global Agenda: National Health Workforce Accounts (NHWA) Regional Perspective and Priorities: evidence, information, and knowledge on human resources for health (HRH) in the Americas	Mathieu Boniol, World Health Organization (WHO) Gisele Almeida, PAHO
10:30–10:45	Coffee break	
10:45–11:30	Human resources for health information systems (HRHIS) in the Region • Brazil • Cuba	Moderator: Isabel Duré, PAHO
11:30–12:30	Regional perspectives: HRH data and information • Overview of user survey responses • Validation of basic indicators for Latin America	Isabel Duré, PAHO Ana Paula Cavalcante de Oliveira, PAHO Mariana Lopes Galante, PAHO
12:30–13:30	Lunch	
13:30–13:50	Registration in the Virtual Campus of Public Health (VCPH) Community of Practice	Johel Díaz, VCPH, PAHO Edgardo de Gracia, VCPH, PAHO

Time	Activity	Speaker/presenter/moderator
13:50–14:50	National health workforce accounts - NHWA implementation (actor participation and data sources) - Version 2 of the NHWA indicators - NHWA data portal	Teena Kunjumen, WHO Moderator: Gisele Almeida, PAHO
14:50–15:10	NHWA: critical issues - Levels of activities - Types of management/financing: public, private, and other sectors - Education and other sectors - Types of facilities: hospitals, community centers, clinics, etc. - Place of birth and place of training	Teena Kunjumen, WHO
15:10–15:30	NHWA: group work on critical issues	PAHO and WHO teams
15:30–15:45	Coffee break	
15:45–16:00	NHWA: group work on critical issues (continued) Comments to the full group	PAHO and WHO teams
16:00–17:30	NHWA platform training - Data entry (online and Excel batch upload) - Data visualization and extraction - Completion and submission of data - Group work by country on the platform	Teena Kunjumen, WHO Mathieu Boniol, WHO Ana Paula Cavalcante, PAHO Mariana Lopes Galante, PAHO

Day 2: 12 September

Time	Activity	Speaker/presenter/moderator
8:30–9:10	Occupation mapping - Chile - Costa Rica	Moderator: Gisele Almeida, PAHO
9:10–10:30	Occupation mapping Methodology	Mario Dal Poz, WHO Collaborating Center
10:30–10:45	Coffee break	
10:45–12:30	Occupation mapping Group work	PAHO and WHO teams
12:30–13:30	Lunch	
13:30–14:10	Mapping of actors - El Salvador - Paraguay	Moderator: Isabel Duré, PAHO

Time	Activity	Speaker/presenter/moderator
14:10–15:00	Mapping of actors - Proposed methodology - Group work (sources of information, information systems, available data, standardized classification, etc.)	Isabel Duré, PAHO
15:00–15:15	Coffee break	
15:15–17:00	Mapping of actors Group work and presentations	Moderator: Ana Paula Cavalcante, PAHO

Day 3: 13 September

Time	Joint activities with all the countries in the Americas	Speaker/presenter/moderator
8:00–8:30	Registration (only for English-speaking participants from the Caribbean, Canada, and United States)	
8:30–9:00	Welcome and opening remarks (participants from the Caribbean, Canada, and United States) Group photo with all participants	Benjamín Puertas, PAHO Leah-Mari Richards, PAHO, Panama
9:00–10:15	HRHIS maturity - Information Systems for Health (IS4H) Maturity Model and conceptual framework - Bahamas - Panama - Uruguay	Andrea Gerger, PAHO Moderator: Gisele Almeida, PAHO
10:15–10:30	VCPH Community of Practice registration	Johel Díaz, PAHO Edgardo de Gracia, PAHO
10:30–10:45	Coffee break	
10:45–12:15	Methodology for HRHIS maturity assessment - Proposed methodology - Group work - Validation	Michele Guillo, PAHO Moderator: Isabel Duré, PAHO
12:15–12:30	Closing for Latin American country focal points	Benjamín Puertas, PAHO
12:30–13:30	Lunch	
13:30–17:30	Questions on the NHPA platform (open to all NHPA focal points) - Data entry on the platform - NHPA portal - Questions	Mathieu Boniol, WHO Teena Kunjumen, WHO Ana Paula Cavalcante, PAHO Mariana Lopes Galante, PAHO

Annex 2.

Agenda of the regional workshop “Development of human resources for health information systems in the Americas” for English-speaking Caribbean countries and territories, Canada, and the United States, 13–15 September 2023

Day 1: 13 September

Time	Joint activities with all the countries in the Americas	Speaker/presenter/moderator
8:00–8:30	Registration (only for English-speaking participants from the Caribbean, Canada, and United States)	
8:30–9:00	Welcome and opening remarks (participants from the Caribbean, Canada, and United States) Group photo with all participants	Benjamín Puertas, Pan American Health Organization (PAHO) Leah-Mari Richards, PAHO, Panama
9:00–10:15	Human resources for health information system (HRHIS) maturity - Information Systems for Health (IS4H) Maturity Model and conceptual framework - Bahamas - Panama - Uruguay	Andrea Gerger, PAHO Moderator: Gisele Almeida, PAHO
10:15–10:30	Registration in the Virtual Campus of Public Health (VCPH) Community of Practice	Johel Díaz, PAHO Edgardo de Gracia, PAHO
10:30–10:45	Coffee break	
10:45–12:15	Methodology for HRHIS maturity assessment - Proposed methodology - Group work - Validation	Michele Guillou, PAHO Moderator: Isabel Duré, PAHO
12:15–12:30	Closing for Latin American country focal points	Benjamín Puertas, PAHO
12:30–13:30	Lunch	
13:30–17:30	Questions on the NHTA platform (open to all NHTA focal points) - Data entry on the platform - NHTA portal - Questions	Mathieu Boniol, World Health Organization (WHO) Teena Kunjumen, WHO Ana Paula Cavalcante, PAHO Mariana Lopes Galante, PAHO

Day 2: 14 September

Time	Activity	Speaker/presenter/moderator
8:30–9:45	Global perspective: NHWA Regional perspective and priorities: evidence, information, and knowledge on human resources for health (HRH)	Mathieu Boniol, WHO Gisele Almeida, PAHO
9:45–10:30	CARICOM's HRH Caribbean Commission Road map of the HRH Caribbean Commission Basic HRH indicators of the HRH Caribbean Commission	Benjamín Puertas, PAHO Helen Royer, CARICOM Fiona Harris-Glenville, PAHO
10:30–10:45	Coffee break	
10:45–11:25	HRHIS • Barbados • Canada	Moderator: Annella Auer, PAHO
11:25–12:30	Regional perspective: human resources data and information • Summary of user survey responses • Analysis of regional indicators	Ana Paula Cavalcante, PAHO Mariana Lopes Galante, PAHO
12:30–13:30	Lunch	
13:30–14:00	NHWA • NHWA implementation (actor participation and data sources) • Version 2 of the NHWA indicators • NHWA data portal	Teena Kunjumen, WHO Moderator: Annella Auer, PAHO
14:00–14:20	NHWA: critical issues • Introduction • Levels of activities • Ownership of facilities: public, private, and other sectors • Education and other sectors • Types of facilities: hospitals, community centers, clinics, etc. • Place of birth and place of training	Teena Kunjumen, WHO Moderator: Ana Paula Cavalcante, PAHO
14:20–14:50	NHWA: group work on critical issues • Comments to the full group	PAHO and WHO teams
14:50–15:30	NHWA training	PAHO and WHO teams
15:30–15:45	Coffee break	
15:45–17:30	NHWA platform training • Data entry (online and Excel batch upload) • Data visualization and extraction • Completion and submission of data • Group work by country on the platform	PAHO and WHO teams

Day 3: 15 September

Time	Activity	Speaker/presenter/moderator
8:30–9:10	Occupation mapping • Jamaica • Suriname	Moderator: Gisele Almeida, PAHO
9:10–10:30	Occupation mapping • Methodology	Mario Dal Poz, WHO Collaborating Center
10:30–10:45	Coffee break	
10:45–12:30	Occupation mapping • Group work and presentations	PAHO and WHO teams
12:30–13:30	Lunch	
13:30–14:10	Mapping of actors • Belize • Guyana	Moderator: Annella Auer, PAHO
14:10–15:00	Mapping of actors • Methodology • Group work (sources of information, information systems, available data, standardized classification, etc.)	Isabel Duré, PAHO
15:00–15:15	Coffee break	
15:15–17:00	Mapping of actors • Group work and presentations	Moderator: Ana Paula Cavalcante, PAHO
17:00	Closing	Gisele Almeida, PAHO Leah-Mari Richards, PAHO, Panama

Annex 3.

Representatives of the various focal points at the regional workshop “Development of human resources for health information systems in the Americas,” Panama, 2023

No.	Country/affiliation	Name	Position/institution
1	Anguilla	Jacqueline Bryan-Niles	Acting Permanent Secretary of Health, Ministry of Health
2	Antigua and Barbuda	Indira Roberts	Health Information Systems Specialist
3	Bahamas	Paul Haven	Vice President of Human Resources, Doctors Hospital Nassau
4	Barbados	Stacie Goring	Director, Planning and Research Unit, Ministry of Health and Welfare
5	Barbados	Dawn Barrow	Director, Information Systems
6	Belize	Lizett Bell	Deputy Director, Health Services; Head of Nursing, Ministry of Health and Welfare
7	Bolivia (Plurinational State of)	Silvia Cuentas Canaza	Advisor, Human Resources Unit, Ministry of Health and Sports
8	Brazil	Décio Santiago da Silva Jr.	Associate Professor, Department of Industrial Engineering, School of Engineering, State University of Rio de Janeiro (UERJ)
9	Brazil	Mario Dal Poz	Full Professor, Institute of Social Medicine, UERJ
10	Brazil	Isabela Cardoso de Matos Pinto	Secretary, Secretariat for Workforce Management and Education in Health (SGTES), Ministry of Health
11	Brazil	Bruno Guimarães de Almeida	Director, Department of Health Workforce Management and Regulation (DEGERTS), SGTES, Ministry of Health
12	Brazil	Marcelo Marques Lima	Consultant, DEGERTS, SGTES, Ministry of Health
13	British Virgin Islands	Dierdra Wheatley-Peters	Director, Human Resources in Health Services
14	Canada	Patrick Cyr	Director, Human Resources for Health Action Task Force, Strategic Policy Branch, Health Canada
15	Canada	Kathy Lee	Director, Human Resources for Health, Data Management Team, Canadian Institute of Health Information
16	Chile	Claudia Godoy	Advisor, Personnel Management Division, Ministry of Health
17	Colombia	Laura Galvis	Colombian Association of Medical Schools
18	Costa Rica	Gloriana Mora Cascante	Head of the Human Resources for Health Unit, Directorate of Health Services, Ministry of Health
19	Cuba	Marcos del Risco del Río	Human Resources for Health Unit
20	Dominica	Vernanda Raymond	Directorate of Health Services, Ministry of Health
21	Dominican Republic	Nuranni Reyes García	Director, Development of Human Resources for Health

No.	Country/affiliation	Name	Position/institution
22	Ecuador	Monserrath López	National Director of Professional Development and Healthcare Careers
23	El Salvador	Paola Alejandra Cea Tenorio	Technical Consultant, Directorate of Human Resources
24	Grenada	Tishanna Smart	Senior Administrative Officer for Human Resources
25	Guatemala	Luis Felipe García Ruano	Directorate of Health Training, Education, and Research
26	Guyana	Helen Royer	Director of Human Development, Caribbean Community
27	Guyana	Rahual Lildhar	Economist, Ministry of Health
28	Haiti	Ketler Guillaume	Director, Directorate of Human Resources for Health, Ministry of Public Health and Population
29	Honduras	Isnaya Nuila Zapata	Director, General Directorate of Development for Human Resources for Health
30	Jamaica	Nastasia Sancko	Digital Health Analyst
31	Jamaica	Stephanie Bernard-Stephens	Health Planning Analyst, Division of Policy, Planning, and Development
32	Montserrat	Arlene Ponteen	Director, Secondary Health Care Services
33	Nicaragua	Christian Amaru Ramos Benavides	Computer Systems Analyst "A," General Directorate of Human Resources, Ministry of Health.
34	Panama	Dagoberto Pino	Head, Human Resources Planning and Research Department
35	Panama	Johel Díaz	Virtual Campus Specialist
36	Panama	Dianne Torres Quintero	Physician, Human Resources Planner, Ministry of Health
37	Paraguay	Cristina Caballero García	Director, Strategic National Directorate for Human Resources for Health
38	Peru	Javier Cesar Loayza Altamirano	Technical Team, Executive Directorate of Planning, Ministry of Health
39	Saint Kitts and Nevis	Kishma Cranstoun Nisbett	Administrative Officer, Ministry of Health
40	Saint Kitts and Nevis	Candida Richards	Administrative Officer, Ministry of Health
41	Saint Lucia	Kevin Harris	Ministry of Health
42	Saint Vincent and the Grenadines	Simone Keizer-Beache	Chief Medical Officer, Ministry of Health, Welfare, and the Environment
43	Saint Vincent and the Grenadines	Neilon Henry	Database Administrator, Ministry of Health, Welfare, and the Environment
44	Saint Vincent and the Grenadines	Whilma Ryan	Second Administrative Officer, Ministry of Health, Welfare, and the Environment
45	Executive Secretariat of the Council of Ministers of Health of Central America and the Dominican Republic (SE-COMISCA)	Melissa Hernández	SE-COMISCA

No.	Country/affiliation	Name	Position/institution
46	Suriname	Muersia Williams	Senior Policy Officer, Ministry of Health
47	Trinidad and Tobago	Sonia Seecharan-Raphael	General Manager, Human Resources, Eastern Regional Health Authority
48	Turks and Caicos Islands	Frankira Forbes	Ministry of Health
49	Uruguay	María Lucía Aicardi	Focal Point, National Health Workforce Accounts, Ministry of Health
50	Venezuela (Bolivarian Republic of)	Eulices Rojas	General Director, Human Resources Office, Ministry of Popular Power for Health
51	World Health Organization (WHO)	Mathieu Boniol	WHO
52	WHO	Teena Kunjumen	WHO
53	Subregional Program for Central America and the Dominican Republic (CAM)	Katiuska Alvayero	CAM National Consultant, Pan American Health Organization (PAHO)
54	PAHO/WHO	Naail Uthman	National Associate, Health Systems Strengthening, Guyana
55	PAHO/WHO	Ana Paula Cavalcante	International Consultant, PAHO
56	PAHO/WHO	Andrea Gerger	Information Systems and Data Management Specialist, Information Systems for Health Unit, Evidence and Intelligence for Action in Health Department
57	PAHO/WHO	Annella Auer	International Expert and Consultant, Pacific Community (SPC) and Bacteriological Reference Center (CRB), PAHO
58	PAHO/WHO	Armando De Negri	Advisor, Health Systems and Services, Mexico
59	PAHO/WHO	Carlos Yuri Ayala Cerna	Advisor, Health Systems and Services, Nicaragua
60	PAHO/WHO	María Fernanda Meneses	National Consultant, Costa Rica
61	PAHO/WHO	Caroline Chang	Advisor, Health Systems and Services, Honduras
62	PAHO/WHO	Cristiane Gosh	National Consultant, Brazil
63	PAHO/WHO	Daniel Albrecht	Advisor, Health Systems and Services, Guyana
64	PAHO/WHO	Benjamin Puertas	Human Resources for Health Unit
65	PAHO/WHO	Evelyn Martínez Cruz	National Consultant
66	PAHO/WHO	Felipe Sánchez	National Consultant, Human Resources for Health, PAHO, Bolivia (Plurinational State of)
67	PAHO/WHO	Fiona Harris-Glenville	International Consultant on Human Resources for Health, Human Resources for Health Unit, PAHO
68	PAHO/WHO	Fredy Canchihuamán	National Consultant, Health Systems and Services, PAHO, Peru
69	PAHO/WHO	Gabriel Listovsky	Virtual Campus for Public Health, PAHO, Panama
70	PAHO/WHO	Gilma Ramos	National Consultant, Health Systems and Services, El Salvador
71	PAHO/WHO	Gisele Almeida	Advisor, Health Systems and Policy Research, PAHO, Panama

No.	Country/affiliation	Name	Position/institution
72	PAHO/WHO	Hernán Sepúlveda	Advisor, Human Resources for Health, Subregional Program for South America, PAHO, Peru
73	PAHO/WHO	Ileana Fleitas	Advisor, Health Systems and Services, Venezuela (Bolivarian Republic of)
74	PAHO/WHO	Isaías Montilla	National Consultant, Health Systems and Services, Panama
75	PAHO/WHO	Javier Uribe	Advisor, Health Systems and Services, Ecuador
76	PAHO/WHO	Joel Caraballo	International Consultant, PAHO
77	PAHO/WHO	Leah-Mari Richards	Advisor, Health Systems and Services, Panama
78	PAHO/WHO	María Isabel Duré	International Consultant, PAHO
79	PAHO/WHO	Mariana Lopes	Intern, PAHO
80	PAHO/WHO	Michele Guillou	International Consultant, Human Resources for Health Unit, PAHO
81	PAHO/WHO	Monica Duraes	National Consultant, Brazil
82	PAHO/WHO	Oristela Canto	Administrative Assistant
83	PAHO/WHO	Rosmond Adams	Advisor, Health Systems and Services, Suriname
84	PAHO/WHO	Rita de Carol	Administrative Assistant
85	PAHO/WHO (Community-Based Rehabilitation [CBR] Network)	Karen Gladbach	International Consultant, PAHO Subregional Program for the Caribbean (SPC-CRB)

Human resources for health information systems (HRHIS) are essential for efficiently managing healthcare personnel and formulating evidence-based policies. However, many countries face challenges in integrating these systems, which hinders workforce planning and monitoring. The fragmentation of data poses a challenge that can be overcome through better coordination with the National Health Workforce Accounts (NHWA).

In September 2023, a workshop on HRHIS was held with more than 90 participants from 35 countries and territories. During the event, significant gaps were identified in the development and governance of these systems, highlighting the need for cooperation tailored to each country's needs. A key topic was the mapping of health occupations, with emphasis on the importance of consistent categorization to optimize the management of human resources.

HRHIS maturity levels were also discussed, and experiences were exchanged on the challenges and opportunities to strengthen these systems. The need to invest in technology and training to meet the growing demands of the sector was highlighted. The mapping of key actors in human resources management was also discussed, with a view to promoting collaboration and enhancing governance.

The workshop fostered cooperation among countries by facilitating the exchange of knowledge and the development of guidance and other materials aimed at supporting the advancement of HRHIS in Latin America and the Caribbean.

PAHO



Pan American
Health
Organization



World Health
Organization
Americas Region

www.paho.org